

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3422

3 Credits: 4

Program Core

Source-grounded

Elements of Food Engineering

Study the official course objectives in the source syllabus.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 3422 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3422.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Food Processing Technology
Course code	3422
Course title	Elements of Food Engineering
Semester	3 Credits: 4
Course category	Program Core
Periods per week	4 (L:3 T:1 P: 0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

12 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Study the official course objectives in the source syllabus.

Course outcomes

CO	Official outcome	Study level
CO1	Understanding properties of food materials. 17 Discuss theoretical knowledge of frictional	See official syllabus
CO2	15 Understanding properties of food materials Distinguish thermal, electrical and electromagnetic	See official syllabus
CO3	15 Understanding properties of the food Explain rheological properties related to	See official syllabus
CO4	Understanding qualitative and quantitative attributes of food. 13 CO-PO Mapping Course PO1 PO2 PO3 PO4 PO5 PO6 PO7 Outcomes	See official syllabus
CO1	2	See official syllabus
CO2	2	See official syllabus
CO3	2	See official syllabus
CO4	2 3-Stronglymapped, 2-Moderatelymapped, 1-Weaklymapped	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
The	The CO-PO mapping is reproduced in the official source PDF where provided.

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Explain physical, aero and hydrodynamic properties of food materials.	3	As specified
Module 2	Discuss theoretical knowledge of frictional properties of food materials	3	As specified
Module 3	Distinguish thermal, electrical and electromagnetic properties of the food	3	As specified
Module 4	attributes of food.	3	As specified

MODULE 1

Explain physical, aero and hydrodynamic properties of food materials.

This module is presented from the official syllabus scope for Elements of Food Engineering. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain physical, aero and hydrodynamic properties of food materials.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ Explain the concept of various physical 6 Understanding properties such as volume, porosity. Describe the various hydrodynamic

Explain the concept of various physical 6 Understanding properties such as volume, porosity. Describe the various hydrodynamic is an official topic in Module 1 of Elements of Food Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the concept of various physical 6 Understanding properties such as volume, porosity. Describe the various hydrodynamic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the concept of various physical 6 understanding properties such as volume, porosity. describe the various hydrodynamic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the concept of various physical 6 Understanding properties such as volume, porosity. Describe the various hydrodynamic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain the concept of various physical 6 Understanding properties such as volume, porosity. Describe the various hydrodynamic definition/meaning .
 , steps, application .
 Technical terms English- .

▼ 5 Understanding properties such as, density, viscosity etc. State the various aerodynamic properties

5 Understanding properties such as, density, viscosity etc. State the various aerodynamic properties is an official topic in Module 1 of Elements of Food Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding properties such as, density, viscosity etc. State the various aerodynamic properties using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding properties such as, density, viscosity etc. state the various aerodynamic properties should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding properties such as, density, viscosity etc. State the various aerodynamic properties means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-5 Understanding properties such as, density, viscosity etc. State the various aerodynamic properties definition/meaning. Steps, application. Technical terms English- Malayalam.

▼ 6 Understanding such as drag coefficient and terminal velocity porosity, viscosity, drag coefficient, terminal velocity.

6 Understanding such as drag coefficient and terminal velocity porosity, viscosity, drag coefficient, terminal velocity. is an official topic in Module 1 of Elements of Food Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding such as drag coefficient and terminal velocity porosity, viscosity, drag coefficient, terminal velocity. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding such as drag coefficient and terminal velocity porosity, viscosity, drag coefficient, terminal velocity. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding such as drag coefficient and terminal velocity porosity, viscosity, drag coefficient, terminal velocity. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 6 Understanding such as drag coefficient and terminal velocity porosity, viscosity, drag coefficient, terminal velocity.
 പരസ്പരം definition/meaning.
 steps, application. Technical terms English-
 .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Discuss theoretical knowledge of frictional properties of food materials

This module is presented from the official syllabus scope for Elements of Food Engineering. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Discuss theoretical knowledge of frictional properties of food materials
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ Describe the laws of friction 5 Understanding Review kinetic friction, static friction, Understanding

Describe the laws of friction 5 Understanding Review kinetic friction, static friction, Understanding is an official topic in Module 2 of Elements of Food Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the laws of friction 5 Understanding Review kinetic friction, static friction, Understanding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the laws of friction 5 understanding review kinetic friction, static friction, understanding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Describe the laws of friction 5 Understanding Review kinetic friction, static friction, Understanding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Describe the laws of friction 5 Understanding Review kinetic friction, static friction, Understanding definition/meaning. Steps, application. Technical terms English- Malayalam.

▼ coefficient of friction, rolling resistance 5 Define angle of repose, angle of internal Understanding

coefficient of friction, rolling resistance 5 Define angle of repose, angle of internal Understanding is an official topic in Module 2 of Elements of Food Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify coefficient of friction, rolling resistance 5 Define angle of repose, angle of internal Understanding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, coefficient of friction, rolling resistance 5 define angle of repose, angle of internal understanding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What coefficient of friction, rolling resistance 5 Define angle of repose, angle of internal Understanding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- coefficient of friction, rolling resistance 5 Define angle of repose, angle of internal Understanding definition/meaning . steps, application . Technical terms English- .

▼ friction 4 of friction, rolling resistance, Definition and concept of angle of repose, angle of internal friction.

friction 4 of friction, rolling resistance, Definition and concept of angle of repose, angle of internal friction. is an official topic in Module 2 of Elements of Food Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify friction 4 of friction, rolling resistance, Definition and concept of angle of repose, angle of internal friction. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, friction 4 of friction, rolling resistance, definition and concept of angle of repose, angle of internal friction. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What friction 4 of friction, rolling resistance, Definition and concept of angle of repose, angle of internal friction. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- ഫ്രിക്ഷൻ 4 ഫ്രിക്ഷൻ, റോളിംഗ് റെസിസ്റ്റൻസ്, ഡിഫിനിഷൻ ആൻഡ് കോൺസെപ്റ്റ് ഓഫ് റിപ്പോസ് ഓംഗിൾ, റോളിംഗ് ഫ്രിക്ഷൻ.

ഫ്രിക്ഷൻ ഫ്രിക്ഷൻ ഡിഫിനിഷൻ/മീനിംഗ് ഫ്രിക്ഷൻ. ഫ്രിക്ഷൻ ഫ്രിക്ഷൻ ഫ്രിക്ഷൻ, steps, application ഫ്രിക്ഷൻ ഫ്രിക്ഷൻ ഫ്രിക്ഷൻ. Technical terms English-ൽ ഫ്രിക്ഷൻ ഫ്രിക്ഷൻ.

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Distinguish thermal, electrical and electromagnetic properties of the food

This module is presented from the official syllabus scope for Elements of Food Engineering. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Distinguish thermal, electrical and electromagnetic properties of the food
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ Describe thermal conductivity 5 Understanding Define specific heat, measurement of

Describe thermal conductivity 5 Understanding Define specific heat, measurement of is an official topic in Module 3 of Elements of Food Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe thermal conductivity 5 Understanding Define specific heat, measurement of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe thermal conductivity 5 understanding define specific heat, measurement of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Describe thermal conductivity 5 Understanding Define specific heat, measurement of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Describe thermal conductivity 5 Understanding Define specific heat, measurement of definition/meaning . . . , steps, application Technical terms English-

▼ specific heat 5 Understanding State thermal diffusivity,

specific heat 5 Understanding State thermal diffusivity, is an official topic in Module 3 of Elements of Food Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify specific heat 5 Understanding State thermal diffusivity, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, specific heat 5 understanding state thermal diffusivity, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What specific heat 5 Understanding State thermal diffusivity, means in the official course context

▼ boiling point, freezing 5 Understanding point measurement, Concepts and definition of thermal diffusivity, boiling point, freezing point. Explain rheological properties related to qualitative and quantitative

boiling point, freezing 5 Understanding point measurement, Concepts and definition of thermal diffusivity, boiling point, freezing point. Explain rheological properties related to qualitative and quantitative is an official topic in Module 3 of Elements of Food Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify boiling point, freezing 5 Understanding point measurement, Concepts and definition of thermal diffusivity, boiling point, freezing point. Explain rheological properties related to qualitative and quantitative using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, boiling point, freezing 5 understanding point measurement, concepts and definition of thermal diffusivity, boiling point, freezing point. explain rheological properties related to qualitative and

quantitative should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What boiling point, freezing 5 Understanding point measurement, Concepts and definition of thermal diffusivity, boiling point, freezing point. Explain rheological properties related to qualitative and quantitative means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-
boiling point, freezing 5 Understanding
point measurement, Concepts and definition of thermal diffusivity, boiling
point, freezing point. Explain rheological properties related to qualitative and
quantitative
 definition/meaning .
 , steps, application
 .
Technical terms English-
 .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4**attributes of food.**

This module is presented from the official syllabus scope for Elements of Food Engineering. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: attributes of food.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ rheology

rheology is an official topic in Module 4 of Elements of Food Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify rheology using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, rheology should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What rheology means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-റൂറ rheology റൂറററററററററററ റൂററററ definition/meaning റൂററററററററററററ. റൂറററററ റൂറററററ റൂററററററ, steps, application റൂറററററ റൂററററററററററ റൂറററററ. Technical terms English-റൂ റൂററററ റൂറററററററററററ.

▼ Classify the rheology of food materials 4 Understanding Illustrate the physical states of

Classify the rheology of food materials 4 Understanding Illustrate the physical states of is an official topic in Module 4 of Elements of Food Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Classify the rheology of food materials 4 Understanding Illustrate the physical states of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, classify the rheology of food materials 4 understanding illustrate the physical states of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Classify the rheology of food materials 4 Understanding Illustrate the physical states of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-
 Classify the rheology of food materials 4
 Understanding Illustrate the physical states of
 definition/meaning .
 , steps,
 application . Technical terms English-
 .

▼ **matter, classic ideal materials 4 Understanding ideal materials.**

TEXT/REFERENCE T/R Book Title/Author Mohensenin N N, Physical

properties of plants animal materials, Gordon and R1 Breach

publishers, New York, 1980 Rao M A , Rizvi S S H,Azim K Datta and

Jasim Ahmed, Engineering properties R2 of foods 4th Ed., CRC Press,

2014 K M Sahay and K K Singh, Unit operations of Agricultural

Processing, Second R3 edition, Vikas Publishing India Pvt Ltd, 2004

Ignacio Arana, Physical Properties of Foods Novel Measurement

Techniques R4 and Applications, 1st edition, CRC Press, 2016

Singhal,O.P. and Samuel,D.V.K, “Engineering Properties of Biological

Materials”. R5 Saroj Prakasan, Allahabad 2003 Peleg, M.and

BagelayE.B., “Physical properties of foods”. AVI publishing Co. USA R6

1983. Online Resources Sl.No WebsiteLink 1

<https://www.khanacademy.org/science/physics/thermodynamics/specific-heat-and-heat-transfer/a/what-is-thermal-conductivity> 2

<http://ecoursesonline.iasri.res.in/mod/page/view.php?id=1013> 3

<https://www.nature.com/articles/eye2017267> 4

<https://acta.fih.upt.ro/pdf/2022-3/ACTA-2022-3-15.pdf>

matter, classic ideal materials 4 Understanding ideal materials.

TEXT/REFERENCE T/R Book Title/Author Mohensenin N N, Physical

properties of plants animal materials, Gordon and R1 Breach publishers,

New York, 1980 Rao M A , Rizvi S S H,Azim K Datta and Jasim Ahmed,

Engineering properties R2 of foods 4th Ed., CRC Press, 2014 K M Sahay and

K K Singh, Unit operations of Agricultural Processing, Second R3 edition,

Vikas Publishing India Pvt Ltd, 2004 Ignacio Arana, Physical Properties of

Foods Novel Measurement Techniques R4 and Applications, 1st edition,

CRC Press, 2016 Singhal,O.P. and Samuel,D.V.K, “Engineering Properties of

Biological Materials”. R5 Saroj Prakasan, Allahabad 2003 Peleg, M.and

BagelayE.B., “Physical properties of foods”. AVI publishing Co. USA R6

1983. Online Resources Sl.No WebsiteLink 1

<https://www.khanacademy.org/science/physics/thermodynamics/specific-heat-and-heat-transfer/a/what-is-thermal-conductivity> 2

<http://ecoursesonline.iasri.res.in/mod/page/view.php?id=1013> 3

<https://www.nature.com/articles/eye2017267> 4

<https://acta.fih.upt.ro/pdf/2022-3/ACTA-2022-3-15.pdf> is an official topic in Module 4 of Elements of Food Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify matter, classic ideal materials 4 Understanding ideal materials. TEXT/REFERENCE T/R Book Title/Author Mohensenin N N, Physical properties of plants animal materials, Gordon and R1 Breach publishers, New York, 1980 Rao M A , Rizvi S S H,Azim K Datta and Jasim Ahmed, Engineering properties R2 of foods 4th Ed., CRC Press, 2014 K M Sahay and K K Singh, Unit operations of Agricultural Processing, Second R3 edition, Vikas Publishing India Pvt Ltd, 2004 Ignacio Arana, Physical Properties of Foods Novel Measurement Techniques R4 and Applications, 1st edition, CRC Press, 2016 Singhal,O.P. and Samuel,D.V.K, “Engineering Properties of Biological Materials”. R5 Saroj Prakasan, Allahabad 2003 Peleg, M.and BagelayE.B., “Physical properties of foods”. AVI publishing Co. USA R6 1983. Online Resources Sl.No WebsiteLink 1 <https://www.khanacademy.org/science/physics/thermodynamics/specific-heat-and-heat-transfer/a/what-is-thermal-conductivity> 2 <http://ecoursesonline.iasri.res.in/mod/page/view.php?id=1013> 3 <https://www.nature.com/articles/eye2017267> 4 <https://acta.fih.upt.ro/pdf/2022-3/ACTA-2022-3-15.pdf> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, matter, classic ideal materials 4 understanding ideal materials. text/reference t/r book title/author mohensenin n n, physical properties of plants animal materials, gordon and r1 breach publishers, new york, 1980 rao m a , rizvi s s h, azim k datta and jasim ahmed, engineering properties r2 of foods 4th ed., crc press, 2014 k m sahay and k k singh, unit operations of agricultural processing, second r3 edition, vikas publishing india pvt ltd, 2004 ignacio arana, physical properties of foods novel measurement techniques r4 and applications, 1st edition, crc press, 2016 singhal, o.p. and samuel, d.v.k, "engineering properties of biological materials". r5 saroj prakasan, allahabad 2003 peleg, m. and bagelaye. b., "physical properties of foods". avi publishing co. usa r6 1983. online resources sl.no websitelink 1 <https://www.khanacademy.org/science/physics/thermodynamics/specific-heat-and-heat-transfer/a/what-is-thermal-conductivity> 2 <http://ecoursesonline.iasri.res.in/mod/page/view.php?id=1013> 3 <https://www.nature.com/articles/eye2017267> 4 <https://acta.fih.upt.ro/pdf/2022-3/acta-2022-3-15.pdf> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What matter, classic ideal materials 4 Understanding ideal materials. TEXT/REFERENCE T/R Book Title/Author Mohensenin N N, Physical properties of plants animal materials, Gordon and R1 Breach publishers, New York, 1980 Rao M A , Rizvi S S H, Azim K Datta and Jasim Ahmed, Engineering properties R2 of foods 4th Ed., CRC Press, 2014 K M Sahay and K K Singh, Unit operations of Agricultural Processing, Second R3 edition, Vikas Publishing India Pvt Ltd, 2004 Ignacio Arana, Physical Properties of Foods Novel Measurement Techniques R4 and Applications, 1st edition, CRC Press, 2016 Singhal, O.P. and Samuel, D.V.K, "Engineering Properties of Biological Materials". R5 Saroj Prakasan, Allahabad 2003 Peleg, M. and Bagelaye E.B., "Physical properties of foods". AVI publishing Co. USA R6 1983. Online Resources Sl.No WebsiteLink 1

Aspect	What to prepare
	https://www.khanacademy.org/science/physics/thermodynamics/specific-heat-and-heat-transfer/a/what-is-thermal-conductivity_2 http://ecoursesonline.iasri.res.in/mod/page/view.php?id=1013_3 https://www.nature.com/articles/eye2017267_4 https://acta.fih.upt.ro/pdf/2022-3/ACTA-2022-3-15.pdf means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ matter, classic ideal materials 4 Understanding ideal materials. TEXT/REFERENCE T/R Book Title/Author Mohensenin N N, Physical properties of plants animal materials, Gordon and R1 Breach publishers, New York, 1980 Rao M A , Rizvi S S H,Azim K Datta and Jasim Ahmed, Engineering properties R2 of foods 4th Ed., CRC Press, 2014 K M Sahay and K K Singh, Unit operations of Agricultural Processing, Second R3 edition, Vikas Publishing India Pvt Ltd, 2004 Ignacio Arana, Physical Properties of Foods Novel Measurement Techniques R4 and Applications, 1st edition, CRC Press, 2016 Singhal,O.P. and Samuel,D.V.K, “Engineering Properties of Biological Materials”. R5 Saroj Prakasan, Allahabad 2003 Peleg, M.and BagelayE.B., “Physical properties of foods”. AVI publishing Co. USA R6 1983. Online Resources SI.No WebsiteLink 1 https://www.khanacademy.org/science/physics/thermodynamics/specific-heat-and-heat-transfer/a/what-is-thermal-conductivity_2
http://ecoursesonline.iasri.res.in/mod/page/view.php?id=1013_3
https://www.nature.com/articles/eye2017267_4

<https://acta.fih.upt.ro/pdf/2022-3/ACTA-2022-3-15.pdf> **Definiție** definiție/meaning **Definiție**. **Definiție** **Definiție** **Definiție**, steps, application **Definiție** **Definiție** **Definiție**. Technical terms English-**Definiție** **Definiție**.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► **Define or explain Explain the concept of various physical 6 Understanding properties such as volume, porosity. Describe the various hydrodynamic. (3 marks)**

► **Define or explain 5 Understanding properties such as, density, viscosity etc. State the various aerodynamic properties. (3 marks)**

► **Define or explain 6 Understanding such as drag coefficient and terminal velocity porosity, viscosity, drag coefficient, terminal velocity.. (3 marks)**

Module 2

► **Define or explain Describe the laws of friction 5 Understanding Review kinetic friction, static friction, Understanding. (3 marks)**

► **Define or explain coefficient of friction, rolling resistance 5 Define angle of repose, angle of internal Understanding. (3 marks)**

► **Define or explain friction 4 of friction, rolling resistance, Definition and concept of angle of repose, angle of internal friction.. (3 marks)**

Module 3

► **Define or explain Describe thermal conductivity 5 Understanding**
Define specific heat, measurement of. (3 marks)

► **Define or explain specific heat 5 Understanding State thermal**
diffusivity,. (3 marks)

► **Define or explain boiling point, freezing 5 Understanding point**
measurement, Concepts and definition of thermal diffusivity, boiling
point, freezing point. Explain rheological properties related to
qualitative and quantitative. (3 marks)

Module 4

► **Define or explain rheology. (3 marks)**

► **Define or explain Classify the rheology of food materials 4**
Understanding Illustrate the physical states of. (3 marks)

► **Define or explain matter, classic ideal materials 4 Understanding ideal**
materials. TEXT/REFERENCE T/R Book Title/Author Mohensenin N N,
Physical properties of plants animal materials, Gordon and R1 Breach
publishers, New York, 1980 Rao M A , Rizvi S S H,Azim K Datta and Jasim
Ahmed, Engineering properties R2 of foods 4th Ed., CRC Press, 2014 K
M Sahay and K K Singh, Unit operations of Agricultural Processing,
Second R3 edition, Vikas Publishing India Pvt Ltd, 2004 Ignacio Arana,
Physical Properties of Foods Novel Measurement Techniques R4 and
Applications, 1st edition, CRC Press, 2016 Singhal,O.P. and
Samuel,D.V.K, “Engineering Properties of Biological Materials”. R5 Saroj
Prakasan, Allahabad 2003 Peleg, M.and BagelayE.B., “Physical
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[https://www.khanacademy.org/science/physics/thermodynamics/specific-](https://www.khanacademy.org/science/physics/thermodynamics/specific-heat-and-heat-transfer/a/what-is-thermal-conductivity)
[heat-and- heat-transfer/a/what-is-thermal-conductivity](https://www.khanacademy.org/science/physics/thermodynamics/specific-heat-and-heat-transfer/a/what-is-thermal-conductivity) 2
<http://ecoursesonline.iasri.res.in/mod/page/view.php?id=1013> 3

<https://www.nature.com/articles/eye2017267> 4

<https://acta.fih.upt.ro/pdf/2022-3/ACTA-2022-3-15.pdf>. (3 marks)

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Explain the concept of various physical 6 Understanding properties such as volume, porosity. Describe the various hydrodynamic.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Describe the laws of friction 5 Understanding Review kinetic friction, static friction, Understanding.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Describe thermal conductivity 5 Understanding Define specific heat, measurement of.**

Module 4

1-mark definition: State the meaning of **rheology.**

3-mark explanation: Explain one principle, process or comparison from

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No WebsiteLink
- <https://www.khanacademy.org/science/physics/thermodynamics/specific-heat-and-heat-transfer/a/what-is-thermal-conductivity>
- <http://ecoursesonline.iasri.res.in/mod/page/view.php?id=1013>
<https://www.nature.com/articles/eye2017267>
- <https://acta.fih.upt.ro/pdf/2022-3/ACTA-2022-3-15.pdf>

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 3422. Verify institutional updates against the current official curriculum.